Robustness of Frozen V-JEPA Representations to Feature Augmentations in Medical Image Segmentation

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1 Problem Definition and Motivation

Medical image segmentation is an important task in medical imaging. It helps doctors identify organs and other anatomical structures in CT scans for diagnosis and treatment planning. Building accurate segmentation models is an important task.

Recent self-supervised vision foundation models, such as V-JEPA 2.1 [4], learn powerful visual representations from large amounts of unlabeled data. These representations can be transferred to downstream tasks using lightweight task-specific models. However, it is still unclear how useful frozen V-JEPA features are for 3D medical image segmentation, especially without fine-tuning the encoder.

In this project, we investigate whether feature augmentations applied directly to frozen V-JEPA embeddings can improve segmentation performance. We use a frozen V-JEPA 2.1 encoder together with a simple point-wise MLP decoder, allowing us to evaluate the quality of the learned representations without changing the pretrained encoder.

The experiments are performed on the Medical Segmentation Decathlon (MSD) spleen and lung datasets [3]. We compare several feature-level augmentation techniques, including Gaussian noise, feature dropout, contrast adjustment, and MixUp. We also investigate different strategies for handling class imbalance on the lung dataset.

Our research question is:

> *Will feature augmentation increase a model's performance and if yes, which ones are the most efficient?*

The objective of this project is to evaluate whether feature-level augmentations can improve segmentation performance when using frozen V-JEPA representations. Success is measured by an improvement in Dice score compared to the baseline model without feature augmentation.

Our main contributions are:

- We build a segmentation pipeline based on a frozen V-JEPA 2.1 encoder and a lightweight MLP decoder.
- We evaluate several feature-level augmentation techniques applied directly to V-JEPA embeddings.
- We compare the performance of different augmentation strategies on spleen and lung segmentation tasks.

- We analyze the strengths and limitations of frozen V-JEPA representations for medical image segmentation.

2 Background and Related Work

Medical image segmentation has been extensively studied using supervised deep learning methods. Encoder–decoder architectures, particularly U-Net and its variants, have become the standard approach due to their ability to combine high-level semantic information with fine spatial details. Among these, nn-UNet [1] is a widely used self-configuring framework that automatically adapts its architecture, preprocessing, and training scheme to a given dataset, and has become a strong standard baseline for biomedical segmentation benchmarks. A subsequent large-scale benchmarking study [2] further showed that many recently proposed architectures claiming to outperform nnU-Net fail to hold up under rigorous, pitfall-free validation, and that a properly configured nnU-Net (optionally scaled to modern hardware) remains a state-of-the-art reference for 3D medical image segmentation. These methods achieve strong performance on many medical imaging benchmarks but typically require large amounts of annotated training data.

To reduce the dependence on manual annotations, self-supervised learning has recently become an important research direction. Instead of learning from labeled examples, self-supervised models are pretrained on large collections of unlabeled images and can later be adapted to downstream tasks. Such models often learn more general visual representations that transfer well across different domains.

V-JEPA (Video Joint Embedding Predictive Architecture) [4] is a recent self-supervised foundation model that learns representations by predicting high-level latent features instead of reconstructing pixels. This predictive objective encourages the model to capture semantic information while avoiding unnecessary low-level details. Although V-JEPA was originally developed for natural videos, its learned representations may also be useful for volumetric medical imaging.

Several recent studies have investigated adapting vision foundation models to medical image analysis. Most existing approaches either fine-tune the pretrained encoder or use more complex decoder architectures for downstream segmentation. However, the performance of completely frozen V-JEPA representations combined with a lightweight decoder remains largely unexplored, especially for three-dimensional CT segmentation.

In this work, we investigate whether frozen V-JEPA 2.1 features contain sufficient information for downstream medical image segmentation. We further analyze the effectiveness of feature-space augmentations and class imbalance handling techniques, providing a detailed evaluation on both spleen and lung tumor segmentation tasks, benchmarked against nn-UNet [1, 2] on the corresponding MSD tasks [3].

3 Methodology

3.1 Overall Pipeline

The objective of this work is to investigate whether representations extracted from a frozen V-JEPA 2.1 encoder [4] can be effectively transferred to three-dimensional medical image segmentation tasks. Instead of fine-tuning the encoder, we use it solely as a feature extractor and train a lightweight segmentation decoder on top of the extracted embeddings.

The overall pipeline is illustrated in Figure 1. Each CT volume is first divided into fixed-size three-dimensional patches and preprocessed to match the input requirements of the pretrained

encoder. The patches are then passed through the frozen V-JEPA encoder to obtain latent feature representations. During training, optional feature-space augmentations are applied directly to these embeddings before they are processed by the segmentation decoder. Finally, the predicted segmentation patches are reconstructed into the final volumetric segmentation masks and evaluated using standard segmentation metrics.

Keeping the encoder frozen allows us to evaluate the transferability of pretrained V-JEPA representations while minimizing the number of trainable parameters. This design also enables a direct comparison of different decoder configurations and feature-space augmentation strategies without modifying the pretrained encoder.

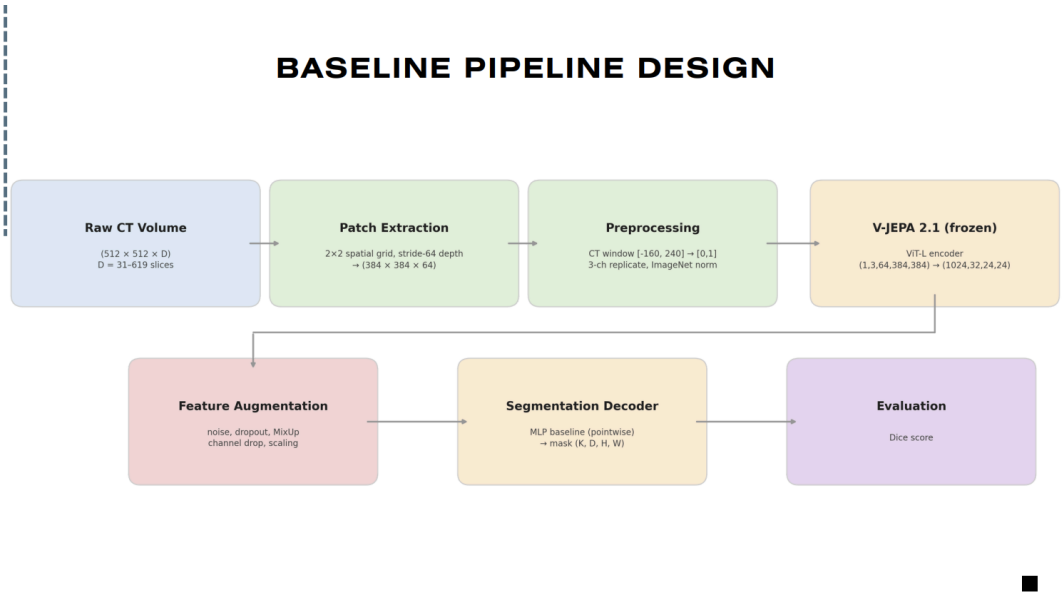


Figure 1: Overview of the proposed segmentation pipeline.

3.2 Datasets

The proposed pipeline was evaluated on two tasks from the Medical Segmentation Decathlon (MSD) [3]: spleen segmentation and lung tumor segmentation. Both datasets consist of three-dimensional computed tomography (CT) volumes together with voxel-level binary segmentation masks.

The spleen dataset contains 41 annotated CT scans, while the lung dataset contains 63 annotated scans. These tasks present different levels of segmentation difficulty. The spleen occupies a relatively large and continuous anatomical structure, whereas lung tumors are considerably smaller, more heterogeneous, and exhibit substantial class imbalance.

To prevent information leakage, all experiments were performed using patient-level data splitting, ensuring that patches extracted from the same CT volume were never shared between the training and validation sets.

3.3 Patch Generation

The original CT volumes differ in spatial dimensions and cannot be processed directly by the pretrained V-JEPA encoder, which requires inputs of a fixed size. Therefore, each volume is divided

into three-dimensional patches before feature extraction.

For every CT scan, overlapping patches of size $384 \times 384 \times 64$ voxels are extracted. Adjacent patches overlap by 50% along each spatial dimension to reduce boundary artifacts and preserve contextual information near patch borders. This strategy also increases the number of training samples while ensuring that neighboring anatomical structures are represented consistently.

Each extracted patch is paired with the corresponding region from the ground-truth segmentation mask, producing aligned image–mask pairs for supervised training. During inference, all predicted patches are reconstructed into the original CT volume by combining the overlapping predictions, resulting in a complete three-dimensional segmentation mask.

Patch-based processing makes it possible to handle high-resolution CT volumes within GPU memory limitations while maintaining sufficient anatomical context for downstream segmentation.

3.4 Preprocessing

Before feature extraction, each CT patch undergoes a series of preprocessing steps to ensure compatibility with the pretrained V-JEPA 2.1 encoder.

First, Hounsfield Unit (HU) intensities are clipped to the range $[-160, 240]$, which emphasizes soft tissue while suppressing irrelevant intensity values. The clipped intensities are then linearly normalized to the interval $[0, 1]$.

Since V-JEPA was pretrained on three-channel natural images, each normalized CT patch is replicated across three identical channels. Finally, ImageNet normalization is applied using the standard channel-wise mean and standard deviation expected by the pretrained encoder.

After preprocessing, each input sample has the shape $(3, 64, 384, 384)$ and is passed to the frozen V-JEPA encoder for feature extraction.

3.5 Frozen V-JEPA Feature Extraction

The preprocessed CT patches are passed through a pretrained V-JEPA 2.1 [4] encoder to obtain high-level latent feature representations. Throughout all experiments, the encoder remains frozen, meaning that its parameters are not updated during training. Only the downstream segmentation decoder is optimized.

For each input patch of shape $(3, 64, 384, 384)$, the encoder produces a feature tensor of shape $(1024, 32, 24, 24)$. These embeddings provide a compact representation of the input volume while preserving high-level semantic information learned during self-supervised pretraining.

Using a frozen encoder significantly reduces the number of trainable parameters and computational cost compared to full end-to-end fine-tuning. More importantly, it allows the quality of the pretrained V-JEPA representations to be evaluated independently of the encoder optimization process. Consequently, any changes in segmentation performance can be attributed to the downstream decoder or feature-space processing rather than modifications of the pretrained backbone.

The extracted feature tensors serve as the input to the segmentation decoder described in the following section.

3.6 Segmentation Decoder

The frozen feature representations extracted by the V-JEPA encoder are processed by a lightweight point-wise multilayer perceptron (MLP) decoder. The decoder is the only trainable component of the proposed pipeline and is designed to evaluate how much segmentation-relevant information is directly decodable from frozen V-JEPA embeddings without modifying the pretrained encoder.

The input to the decoder is a feature tensor of shape $(B, 1024, 32, 24, 24)$, where B denotes the batch size. First, the channel dimension is moved to the last axis and the spatial dimensions are flattened, transforming the feature volume into a sequence of feature vectors. Each spatial location (token) is therefore represented by an independent 1024-dimensional embedding.

A shared point-wise MLP is then applied independently to every feature vector. The decoder consists of three fully connected layers with hidden dimensions of 512 and 128 units, respectively, using the Gaussian Error Linear Unit (GELU) activation function after each hidden layer. The final linear layer produces a single logit corresponding to the foreground probability at each spatial location.

After all tokens have been processed, the predicted logits are reshaped back into a volumetric representation of size $(B, 1, 32, 24, 24)$. Finally, parameter-free trilinear interpolation is used to upsample the low-resolution prediction to the original patch resolution of $(64, 384, 384)$.

Unlike conventional segmentation decoders based on convolutional layers, skip connections, or attention mechanisms, the proposed decoder performs no spatial feature mixing before prediction. Each token is processed independently using the same shared MLP. This intentionally simple architecture allows the segmentation performance to primarily reflect the quality of the frozen V-JEPA representations rather than the capacity of the decoder itself.

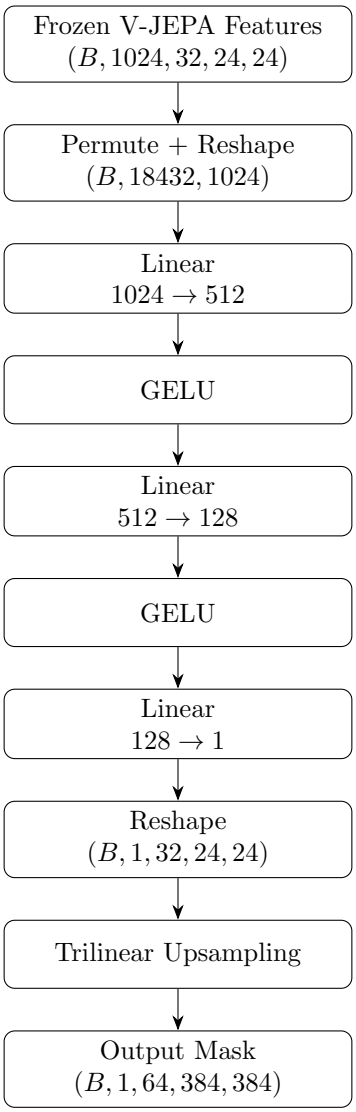


Figure 2: Architecture of the proposed point-wise MLP segmentation decoder.

3.7 Feature-space Augmentation

To investigate whether frozen V-JEPA representations can be made more robust, several feature-space augmentation techniques were applied directly to the extracted embeddings during training. Unlike conventional image augmentations, which modify the input CT images before feature extraction, the proposed methods operate on the latent feature representations produced by the frozen encoder.

Each augmentation was evaluated independently to analyze its influence on downstream segmentation performance. The following feature-space transformations were considered:

- **Gaussian noise**, where random noise sampled from a Gaussian distribution was added to the feature embeddings.
- **Feature dropout**, which randomly sets individual feature values to zero to encourage robustness against missing feature activations.
- **MixUp**, where feature embeddings and their corresponding target masks were linearly interpolated to generate additional training samples.
- **Contrast adjustment**, which modifies the distribution of latent feature values by increasing or decreasing their contrast.

Since the V-JEPA encoder remains frozen throughout training, all augmentation techniques affect only the extracted feature representations and do not alter the pretrained model itself. This design enables the influence of feature-space perturbations to be evaluated independently of encoder optimization.

4 Experimental Setup

4.1 Training Configuration

All experiments used the same training settings to make the comparison fair. The V-JEPA 2.1 encoder was kept frozen during all experiments, and only the MLP decoder was trained. This allowed us to evaluate the quality of the pretrained V-JEPA features without changing the encoder.

The model was trained for 200 epochs without early stopping. We used the Adam optimizer with a learning rate of 0.001 and a batch size of 4. The learning rate was automatically reduced using the ReduceLROnPlateau scheduler with a factor of 0.5 and a patience of 3 epochs when the validation performance stopped improving.

The loss function was a combination of Binary Cross-Entropy (BCE) loss and Dice loss with equal weights. BCE helps classify each voxel as foreground or background, while Dice loss improves the overlap between the predicted mask and the ground-truth mask. The total loss was calculated as:

$$\text{Loss} = 0.5 \times \text{BCE} + 0.5 \times \text{Dice}. \tag{1}$$

The best model was selected using the highest validation Dice score computed on reconstructed CT scans. During training, we recorded the training loss, validation loss, validation Dice score, learning rate, and other training statistics after every epoch to monitor the learning process.

To avoid data leakage, we used GroupKFold with five patient-level splits. This ensured that patches from the same CT scan were never used in both the training and validation sets. Because of limited computing resources, only the first fold was used in all experiments.

4.2 Evaluation Protocol

The proposed pipeline was evaluated on both spleen and lung tumor segmentation tasks using patient-level data splitting. This protocol ensures that all patches originating from the same CT volume belong exclusively to either the training or validation set, preventing information leakage between the two sets.

Segmentation performance was evaluated on reconstructed three-dimensional volumes rather than individual patches. This provides a more realistic assessment of model performance in practical medical imaging scenarios.

The same evaluation procedure was applied consistently across all experiments to enable a fair comparison between the baseline model, feature-space augmentation methods, and class imbalance handling techniques.

4.3 Compared Experiments

Several experiments were conducted to evaluate different aspects of the proposed segmentation pipeline.

First, a baseline model using frozen V-JEPA features and the proposed MLP decoder was established. Next, multiple feature-space augmentation techniques were evaluated independently to determine their influence on segmentation performance.

Additional experiments investigated different strategies for handling severe class imbalance in lung tumor segmentation, including weighted loss functions and oversampling techniques. Finally, the behavior of the proposed pipeline was analyzed separately on spleen and lung tumor datasets in order to compare its performance on segmentation tasks with different levels of difficulty, using nn-UNet [1, 2] as the reference baseline.

5 Results and Conclusions

5.1 Result Table

Table 1 summarizes the validation Dice score, validation loss, and improvement over the baseline for each feature-space augmentation configuration evaluated on the spleen segmentation task, together with a reference nn-UNet [1] baseline.

Table 1: Validation results for the baseline model, each feature-space augmentation configuration, and the reference nn-UNet [1] baseline.

Experiment	**Augmentation params**	**Val Dice**	**Val Loss**	**Δ vs Baseline**
Baseline (no augmentation)	–	0.9197	0.0461	–
Feature Dropout	$p_{\text{channel}} = 0.10$, $p = 0.5$	0.9246	0.0436	0.0049
Contrast	$\delta = 0.2$, $p = 0.5$	0.9224	0.0447	0.0027
Dropout + Contrast	dropout(0.10, 0.5) + contrast(0.2, 0.5)	0.9256	0.0428	0.0059
MixUp (strong)	$\alpha = 0.7$, $p = 0.8$	0.9228	0.0474	0.0031
MixUp (mild)	$\alpha = 0.4$, $p = 0.5$	0.9238	0.0451	0.0041
All four (noise + dropout + contrast + mixup)	noise(0.05, 0.5) + dropout(0.10, 0.5) + contrast(0.2, 0.5) + mixup(0.4, 0.5)	0.9273	0.0437	0.0076
nn-UNet Baseline [1]	–	0.956	–	0.037

Table 2: Lung Segmentation (Task06, MSD [3]) — Experiment Results

Experiment	**Val Dice**	**Val Loss**	**Extra Params**	**Notes**
Baseline (no oversampling)	0.5557	0.2367	–	full-scan, fold 0
Oversampling ($0.3 \rightarrow 0.5$)	0.5508	0.2460	–	full-scan, fold 0
Weighted loss + oversampling	0.3932	0.3245	pos_weight = 1841.5	full-scan, fold 0
nn-UNet baseline [1]	0.56	–	IoU = 0.45	best epoch eval

5.2 Baseline Performance

The baseline model showed that frozen V-JEPA features contain useful information for medical image segmentation. The model achieved better results on the spleen dataset than on the lung dataset. This was expected because the spleen is a large organ with a clear shape, while lung tumors are much smaller and more difficult to segment.

5.3 Class Imbalance Handling

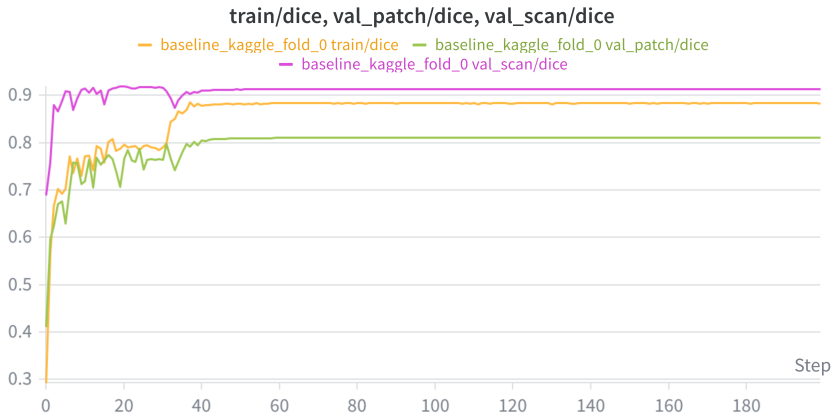
Since lung tumor segmentation is characterized by a severe imbalance between foreground and background voxels, additional experiments investigated methods for mitigating this imbalance during training.

Two commonly used approaches were evaluated: weighted loss functions and oversampling of tumor-containing patches. These methods were compared with the baseline configuration to determine whether emphasizing minority-class samples improves segmentation performance.

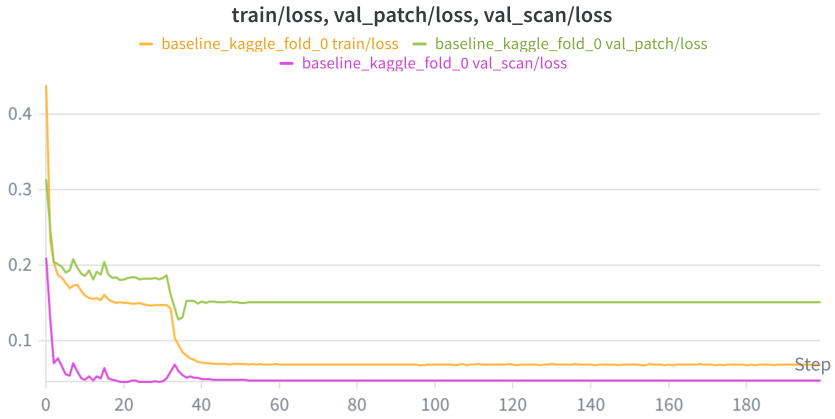
In addition to quantitative evaluation, representative prediction examples were visually inspected to better understand the behavior of the proposed segmentation pipeline. Ground-truth masks were compared with the predicted segmentations to identify correctly segmented regions as well as common failure cases.

These visual comparisons complement the quantitative evaluation by illustrating the strengths and limitations of the proposed approach on both spleen and lung tumor segmentation tasks.

5.4 Visualisations

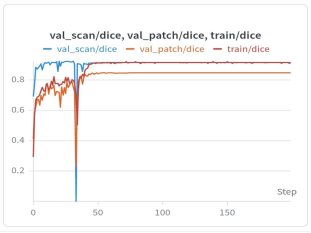


(a) Baseline: Dice/epoch (spleen)

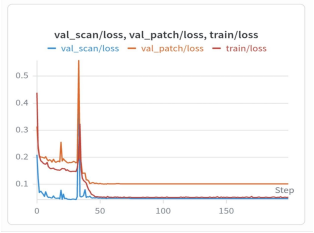


(b) Baseline: Loss/epoch (spleen)

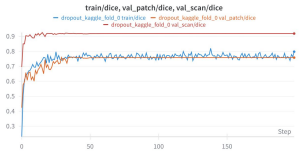
Figure 3: Frozen V-JEPA [4] + MLP decoder baseline (no augmentation) training curves on the MSD spleen task [3].



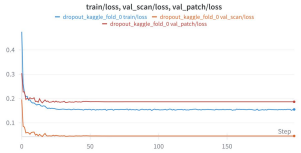
(a) Noise: Dice/epoch



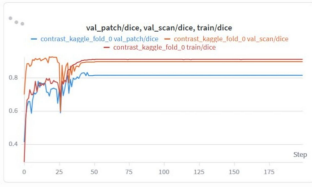
(b) Noise: Loss/epoch



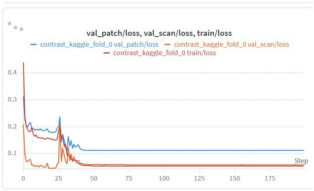
(c) Dropout: Dice/epoch



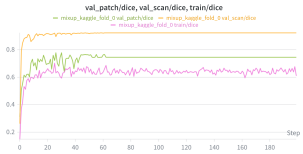
(d) Dropout: Loss/epoch



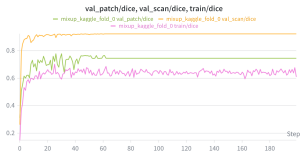
(e) Contrast: Dice/epoch



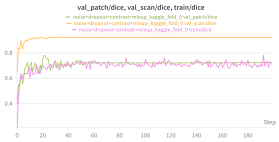
(f) Contrast: Loss/epoch



(g) MixUp (strong): Dice/epoch



(h) MixUp (strong): Loss/epoch

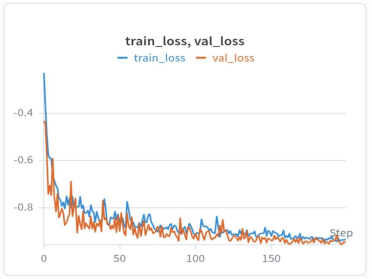


(i) All 4 combined: Dice/epoch



(j) All 4 combined: Loss/epoch

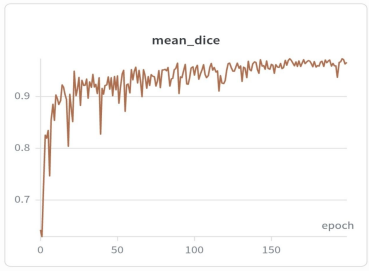
Figure 4: Training curves for each individual feature-space augmentation (noise, dropout, contrast, MixUp) and for all four combined, applied to frozen V-JEPA [4] embeddings on the MSD spleen task [3].



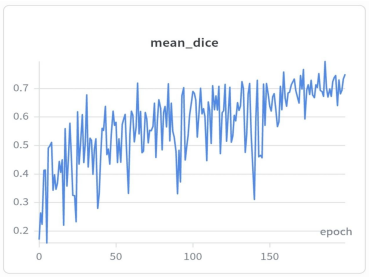
(a) nn-UNet: Spleen loss/epoch



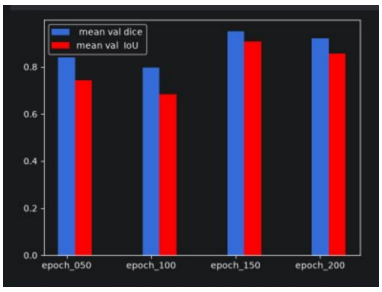
(b) nn-UNet: Lung loss/epoch



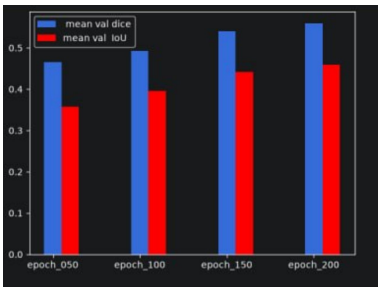
(c) Spleen nn-UNet: mean Dice



(d) Lung nn-UNet: mean Dice

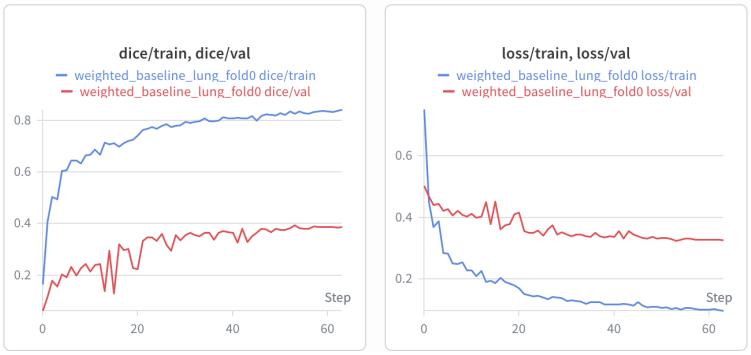


(e) Spleen nn-UNet: validation full scan dice + IoU



(f) Lung nn-UNet: validation full scan dice + IoU

Figure 5: Reference nn-UNet [1, 2] training curves, mean-Dice comparison, and qualitative segmentation examples on both MSD tasks [3].



(a) Lung nn-UNet baseline: loss/epoch



(b) Lung nn-UNet: oversampled + weighted loss

Figure 6: Training loss curves for the lung segmentation experiments.

5.5 Conclusions

In conclusions for spleen task we encountered a consistent very small improvement of dice and loss as a result of augumentations (mean gain +0.0047 Dice), significance is questionable numerically and methodologically, which will be discussed in last segment. In comparison to nn-UNet [1] baseline both V-JEPA with augmentations and without performed worse on 200 folds, athough encountered faster convergence compared to nn-UNet (metrics on our 10th step are comparable to those nn-UNet achieves by 20th). On the lung task, an issue with feasibility of task was encountered - only 0.56 on both nn-UNet and V-JEPA baseline so it was decided to drop task due to problematic extraction of coherent results.

5.6 Limitations and Future Work

This project has several limitations. At the beginning of the project, we trained the baseline model using five-fold cross-validation for 50 epochs. Later, after improving the training pipeline, the final experiments were trained for 200 epochs using only one fold. Because of this, the final results should be confirmed using all folds. Another limitation is the simple MLP decoder. It processes each feature independently and does not use spatial information. Because of this, it cannot fully use the information in the V-JEPA features, and better decoder architectures may achieve higher segmentation accuracy. Feature-level augmentations were evaluated only on the spleen dataset. We did not perform augmentation experiments on the lung dataset because of the higher complexity of the task. Therefore, our conclusions are based only on the spleen results. During this project, we also learned that building a strong baseline is very important before testing new ideas. Better training monitoring and more detailed logging helped us find optimization problems and improve the training pipeline. In future work, we plan to use the existing nnU-Net [1, 2] validation pipeline for a more standardized evaluation of our method. We also want to test the proposed approach on other segmentation tasks, organs, and medical imaging datasets to better understand its generalization ability. Another direction is to replace the simple MLP decoder with more advanced segmentation heads that can use spatial context, such as CNN-based or transformer-based architectures. We also plan to perform full cross-validation to obtain more reliable results. Finally, we would like to investigate multi-scale feature extraction, which may provide richer feature representations and improve segmentation accuracy.

References

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